



Fix by Swapping

Quick facilitation notes & answer key

Goal

Students fix a bug where one block points the wrong way by swapping its direction (forward ↔ back), then run the corrected code to confirm it reaches the target. Altering an existing block to correct the outcome, and describing the effect, is C3.2; the corrected block is a small C3.1 write.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 so students feel Bit move the wrong way, then re-walk the fix.

Run it (about 20 min)

1. I DO: run Program 1 — 'start 1, forward 2 to 3, back 1 to 2. Target is 4 — the back 1 sent Bit the wrong way.'
2. WE DO: Exercise 1 — circle 'back 1'. Exercise 2 — swap it to 'forward 1'; the code becomes 1 → 3 → 4.
3. YOU DO: Exercise 3 — Program 2 lands on 3; swap 'forward 1' to 'back 1' so 3 → 2 → 1.
4. Connect: 'A wrong-way block is fixed by swapping its direction — then run to check.'

Answer key (modelled by the run-gate)

Ex 1 — Program 1: 1 → forward 2 → 3 → back 1 → 2. The wrong-way block is back 1.

Ex 2 — swap back 1 to forward 1: 1 → 3 → 4. Bit reaches the target, 4.

Ex 3 — Program 2: 3 → forward 1 → 4 → back 1 → 3 (misses 1). Swap forward 1 to back 1: 3 → 2 → 1. Bit reaches the target, 1.

Watch for & differentiate

Common slip: changing the block's NUMBER instead of its direction — this bug is about which way it points, so swap forward/back and keep the number.

Simplify: walk Program 1 on the floor; the backward step toward the wrong end is easy to feel.

Extend: ask whether changing a different block could also reach the target, and test the idea.

Success indicator: the child circles the back/forward block that goes the wrong way, swaps it, and reaches the target in both programs.